

Bone Scintiscanning in Osteolytic Lesions

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Abstract: Osteolytic lesions seen on plain radiographs can be caused by various disorders of the bones such as simple bone cyst, aneurysmal bone cyst, plasmacytoma, giant cell tumor, eosinophilic granuloma and tuberculosis. We studied prospectively Tc-99m-methylene diphosphonate bone scan findings in osteolytic lesions seen radiologically and followed them to histopathology. Interestingly, the scans in these patients helped to show if the lesions were monoostotic or polyostotic and, in some cases, ruled out malignant or infective etiology.

Key Words: osteolytic, fibrous dysplasia, bone cyst, eosinophilic granuloma, plasmacytoma, tuberculosis, aneurysmal bone cyst, giant cell tumor, Paget's disease, hyperparathyroidism, multiple myeloma

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Osteolytic lesions of the skeleton are frequently seen in clinical practice. There are several tests available to investigate such lesions. Each one has its advantages and limitations. Bone scintiscanning is important in several aspects in evaluation of these lesions. It is the only test to define the physiology of bones and can also pick up systemic diseases, as illustrated in our cases reported here. Although a bone scan is not specific to diagnose the disease entity itself, it provides important guidelines to further investigations

MATERIALS AND METHODS

Ten patients who presented with osteolytic lesions of the skeleton had bone scans performed using Tc-99m methylene diphosphonate (MDP). Twenty mCi of Tc-99m MDP was used in these cases. All patients were imaged using a low-energy high-resolution collimator on a Millennium MPR gamma camera from Elgems.

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Parathyroid scans were performed using Tc-99m pertechnetate-Tc-99m methoxy-isobutyl-isonitrile (MIBI) subtraction protocol. Initially, a thyroid scan was performed using 2 mCi Tc-99m pertechnetate. Static images of the thyroid gland were obtained in the anterior view at 20 minutes postinjection using a low-energy high-resolution collimator. Twenty mCi Tc-99m MIBI was injected without moving the patient, and sequential static images were acquired at a rate of 5 minutes per frame for 30 minutes. Subtraction of the 2 images was done to obtain the parathyroid image. Histologic evaluation of all the lesions was done except in the case of fibrous dysplasia in view of the characteristic findings seen on bone scans.

Aneurysmal Bone Cyst

A 20-year-old man reported pain in the heel. There was no history of trauma or fever. Plain x-ray revealed a well-defined cystic lesion in the calcaneus (Fig. 1). Bone scan showed hypervascularity in the first phase (Fig. 2) with intense tracer uptake in the third phase (Fig. 3). Histology confirmed the diagnosis of aneurysmal bone cyst.

Aneurysmal bone cysts usually occur in the metaphysis of long bones in the first 2 decades of life. The entity was initially described by Jaffe and Lichtenstein in 1940.¹ In 238 patients with aneurysmal bone cysts studied in the Mayo Clinic files, more than 80% of the lesions were in long bones, flat bones, or the vertebral column.² The cause of aneurysmal bone cysts is unclear. However, some cases with a familial link have been reported.³ Although considered benign, there are anecdotal case reports of malignant transformation of aneurysmal bone cysts.⁴ Hence, some authors have recommended radiotherapy in recurrent tumors.⁵

Eosinophilic Granuloma

A 19-year-old boy presented with pain in the left side of the ribs anteriorly. Plain radiographs revealed an osteolytic lesion in the left fifth rib anterolaterally (Fig. 4). Bone scans showed a central photon-deficient lesion surrounded by a zone of increased tracer uptake (Fig. 5). There was no evidence of involvement of the distant skeleton. Histologic evaluation confirmed the diagnosis of eosinophilic granuloma.

Eosinophilic granuloma is a benign disease of unknown etiology. It is considered to be a part of Langerhans cell



FIGURE 1. Plain X ray of the foot shows well defined osteolytic lesion in the calcaneus.

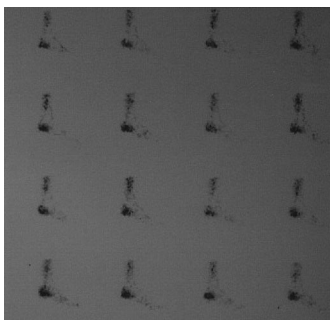


FIGURE 2. First phase shows increased vascularity corresponding to the osteolytic lesion see radiologically.



FIGURE 3. Third phase shows intense localization of tracer in the calcaneus.

histiocytosis. Focal areas of bone are destroyed, which could be solitary or multiple.⁶ Visceral involvement is seen more often before 2 years of age.⁷ After healing, local recurrence can occur as long as 13 years later.⁸

Tuberculosis

A 24-year-old man presented with pain and a discharging sinus in his right foot. Plain radiographs revealed an osteolytic lesion in the proximal end of the right third metatarsal bone (Fig. 6). Bone scans showed hypervascularity in the first phase (Fig. 7) and abnormal tracer localization in the cortical phase (Fig. 8). Histology revealed tuberculosis of the metatarsal. Tuberculosis of the metatarsal bone has been reported before.⁹ Tuberculosis of the skeleton can affect all the bones in the body. Only one third of patients with skeletal tuberculosis could have involvement of the lungs.¹⁰ The

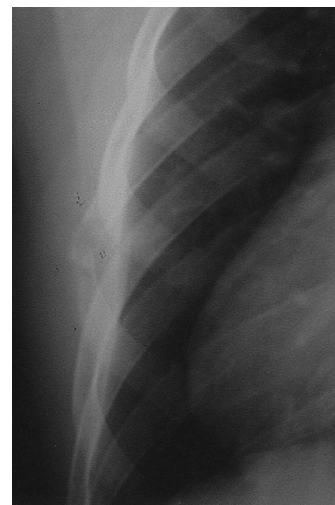


FIGURE 4. X ray of ribs shows osteolytic area in the left fifth rib laterally.



FIGURE 5. Bone scan shows central photon deficiency surrounded by a zone of increased tracer uptake.



FIGURE 6. Plain radiograph of the foot shows osteolytic lesion in the third metatarsal.

incidence of tuberculosis of bone is likely to increase because of the human immunodeficiency virus.¹¹

Musculoskeletal involvement of tuberculosis can mimic common rheumatic disorders. Five syndromes have been described: spondylitis (Pott's disease), peripheral arthritis, osteomyelitis, tenosynovitis, and Poncet's disease (aseptic



FIGURE 7. First phase of bone scan shows increased perfusion in the right foot.

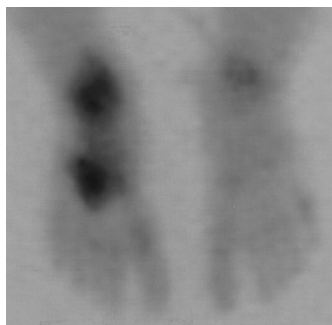


FIGURE 8. Third phase of bone scan shows increased tracer localization in the right third metatarsal bone.

inflammatory polyarthritis in the presence of active tuberculosis, usually involving the lungs).¹²

Giant Cell Tumor

A 30-year-old man presented with painful swelling in the right wrist region. Plain radiographs showed an osteolytic lesion in the lower end of the right ulna (Fig. 9). A bone scan revealed intense uptake corresponding to the osteolytic lesion (Fig. 10). Biopsy confirmed the diagnosis of a giant cell tumor. The swelling recurred after a few months and was reoperated with a wide excision.



FIGURE 9. Plain radiograph of the wrist shows osteolytic lesion in the lower end of right Ulna.



FIGURE 10. Bone scan shows intense localization of tracer in the lower end of right Ulna, corresponding to the lytic lesion seen radiologically.

Giant cell tumor is a locally invasive tumor with a preponderance to recur.¹³ It usually occurs between 20 and 40 years of age, 50% of the time around the knee joint.¹⁴ In the long bones, giant cell tumor usually presents as an eccentric lytic lesion, extending to the subchondral region, with minimal periosteal reaction and an absent sclerotic rim, and often recurs after curettage. Lesions, which can mimic giant cell tumor, are aneurysmal bone cyst, fibrous dysplasia, eosinophilic granuloma, and brown tumor of hyperparathyroidism.¹³

Fibrous Dysplasia

A 22-year-old woman presented with pain in the left hip region. Plain radiographs revealed an osteolytic lesion in the upper shaft of the left femur (Fig. 11). A whole-body bone scan revealed diffusely increased tracer uptake in the shaft of the left femur, the left tibia, the left radius, and the left humerus (Fig. 12). Features were quite suggestive of fibrous dysplasia radiologically.

When there is a long lytic lesion in a long bone, one should think of fibrous dysplasia.¹⁵ Fibrous dysplasia is a disorder of the skeleton, which causes expansion of the bones



FIGURE 11. Plain radiograph of the left hip shows a large osteolytic lesion in the upper shaft of left femur.



FIGURE 12. Whole body bone scan shows diffuse increased tracer uptake in the left humerus, the left radius, left tibia and the left femur. The pattern is characteristic of Fibrous dysplasia.

because of abnormal development of fibrous or connective tissue within the bones. The lesion can be monoostotic or polyostotic.

Fibrous dysplasia is caused by a somatic-activating mutation of the Gs alpha subunit of protein G, resulting in an increased cAMP concentration and resulting in abnormalities of osteoblast differentiation. These osteoblasts produce abnormal bone. There is also an increase in interleukin-6-induced osteoclastic bone resorption, which is the rationale for treating these patients with bisphosphonates. In the past 10 years, bisphosphonate pamidronate has been used by infusion for fibrous dysplasia (2 courses per year) with good results with respect to pain and, in approximately 50% of patients, the refilling of osteolytic lesions.¹⁶



FIGURE 13. Plain radiograph of the pelvis shows osteolysis of the upper end of right femur.

Metastasis

An 82-year-old man presented with pain in the region of the hip. Plain radiographs revealed an osteolytic lesion in the upper end of the right femur (Fig. 13). A 3-phase bone scan revealed an abnormality in all 3 phases (Fig. 14). In addition to this, there was abnormal tracer concentration in the lumbar vertebrae and the ribs, which demonstrated polyostotic involvement suggestive of metastases (Fig. 15). Histology of the femoral lesion confirmed the diagnosis of metastatic adenocarcinoma.

Skeletal metastasis accounts for more than half of malignant bone tumors. The tumors that commonly metastasize to bone are prostate, breast, thyroid, renal, and lung. The skeleton can be involved by direct spread or as a result of hematogenous spread. The bone scan is more sensitive than plain radiographs for detection of metastases.¹⁷ The incidence

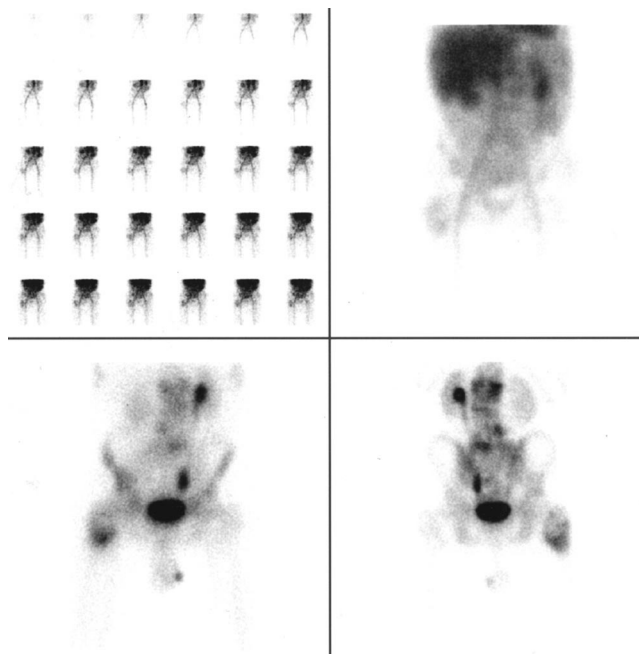


FIGURE 14. Three phase bone scan shows increased perfusion (1st phase), high soft tissue activity (2nd phase) and increased tracer uptake (3rd phase) corresponding to the lytic lesion seen on Plain radiograph.

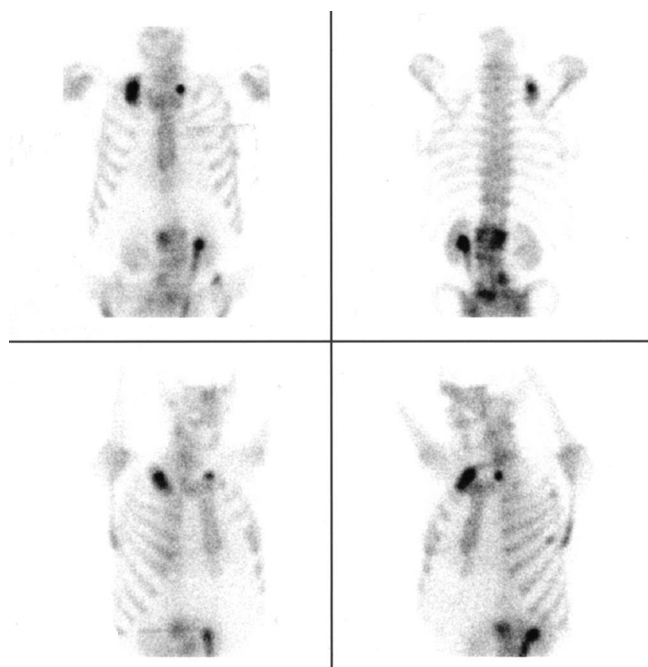


FIGURE 15. Whole body bone scan revealed hot spots in vertebrae and ribs suggesting metastases.

of false-negative bone scans for a metastasis is 3%.¹⁸ Approximately 2% of metastases can be cold on a bone scan. This is commonly seen in multiple myeloma and renal cell carcinoma.¹⁹

Paget's Disease

A 75-year-old woman reported pain in the arm after trivial trauma. Plain radiographs showed a large lytic lesion in the entire shaft of the right humerus (Fig. 16). A bone scan revealed diffusely increased tracer uptake in the entire axial and appendicular skeleton suggestive of Paget's disease (Fig. 17). The mandibular uptake pattern was indicative of Lincoln sign. Her serum alkaline phosphatase was raised and a biopsy from the humerus confirmed the diagnosis of Paget's disease.

Paget's disease of bone is of unknown etiology, can affect any bone, usually occurs after 40 years of age, and can be mono- or polyostotic, involving the skull, femur, tibia, pelvis, and the humerus.²⁰ Involved bones are vascular, weak, and deformed. Men are affected twice as often as women, and it can be seen in 3% of the population.²¹ It is usually discovered incidentally during radiology studies or incidentally discovered as a result of a raised serum alkaline phosphatase.

Histologically, it is characterized by osteoclastic resorption of normal bone by giant multinucleated cells. There is hypervascular osteoblastic activity resulting in distortion of bone matrix.²²



FIGURE 16. Plain radiograph reveals a large osteolytic lesion involving the entire shaft of right humerus. Initially suspected as metastasis.

Hyperparathyroidism

A 52-year-old woman started with pain in the ankle region and shin of 4 years' duration. There was no history of trauma. The symptoms responded to analgesics initially but recurred in a few months. Recently, she started to limp because of pain in the pelvis. Plain radiographs of the tibia revealed a lytic lesion in a small portion of the shaft (Fig. 18). There was generalized osteoporosis.

A whole-body bone scan showed typical features of metabolic bone disease (Fig. 19). The serum calcium was 11.8 (normal, 9–11 mg/dL), serum phosphorous 5.8 mg/dL (normal, 2.5–4.5 mg/dL), and serum alkaline phosphatase 1333 units/L (normal, 98–279 units/L).

Parathyroid scan performed with Tc-99m MIBI (methoxy-isobutyl isonitrile) showed a parathyroid adenoma in the lower pole region of the left lobe of the thyroid (Fig. 20). Histopathology confirmed the diagnosis.

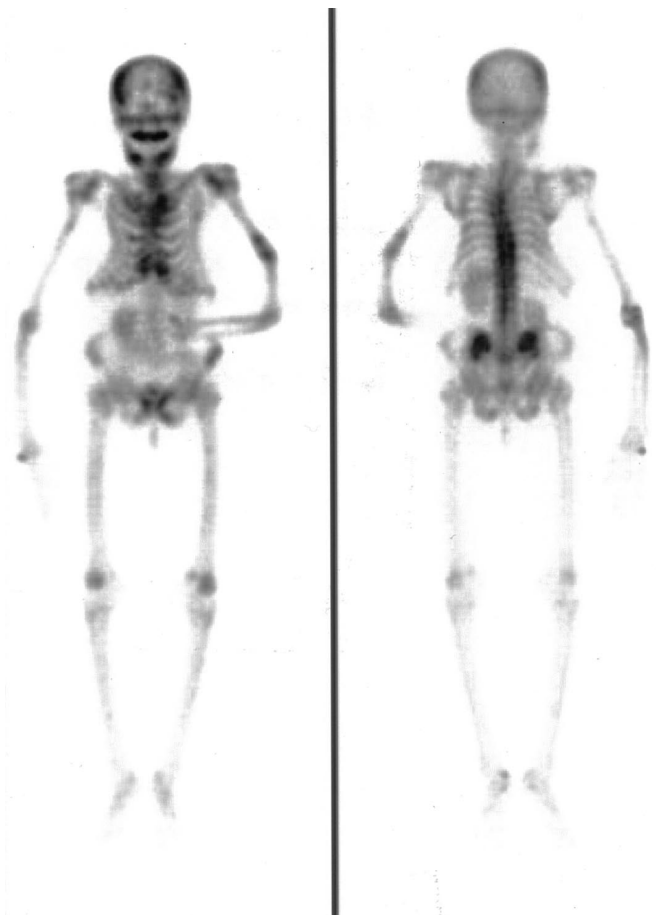


FIGURE 17. Whole body bone scan shows no obvious evidence of skeletal metastasis. Diffuse increased tracer uptake and "the Lincoln sign" suggestive of Paget's disease.

Hyperparathyroidism is a disease often recognized by renal stones, abdominal groans, psychic moans, and ions.²³ Patients with bilateral or recurrent renal calculi, hypercalcemia, and osteoporosis should be investigated for hyperparathyroidism. Hyperparathyroidism is characterized by hypersecretion of parathyroid hormone either because of an adenoma or secondary to hyperplasia of the glands. Parathyroid adenomas can be familial or part of multiple endocrine neoplasia. Hypercalcemia is frequently discovered accidentally.

Hyperparathyroidism causes loss of cortical bone and an increase of trabecular bone. The principal laboratory abnormality in hyperparathyroidism is hypercalcemia. In hyperproteinemia, the total serum calcium can be high but the ionized calcium remains normal, whereas in primary hyperparathyroidism, the ionized calcium is always elevated. There is excess urinary loss of phosphates in the presence of low or low-normal serum phosphorous. The urinary calcium excretion can be high, but it is often low for the degree of hypercalcemia. The alkaline phosphatase is elevated and a high parathyroid



FIGURE 18. Plain radiograph of the right tibia shows a small osteolytic lesion medially.

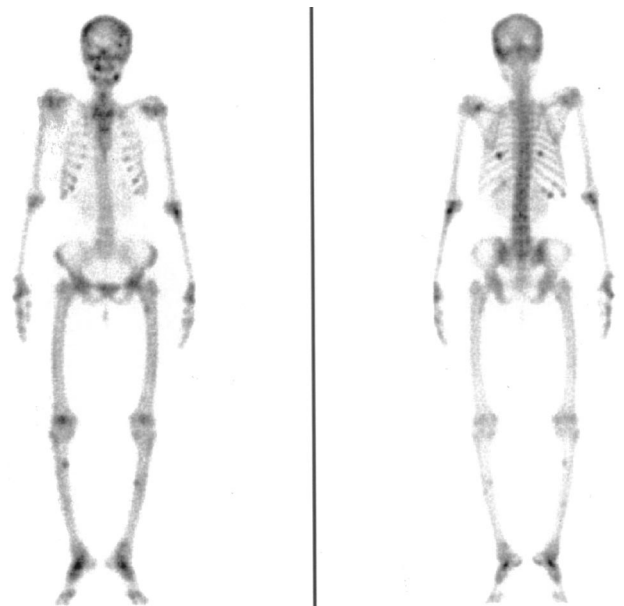


FIGURE 19. Whole body bone scan shows diffuse increased tracer uptake in the entire axial and appendicular skeleton, no visualization of the kidneys and hot spots in the ribs which is characteristic of Metabolic bone disease.

hormone level is an invariable finding. Localization of parathyroid adenomas can be done with Tc-99m pertechnetate-Tc-99m-MIBI subtraction imaging, which has a sensitivity of



FIGURE 20. Parathyroid scan shows left lower pole adenoma.



FIGURE 21. Plain X ray shows lytic lesion with well defined margins in the upper end of left tibia.

87% and a specificity of 95%. Sonography of the neck has a sensitivity of 80%.

Simple Bone Cyst

A 16-year-old boy presented with occasional pain in the left knee region. There was no history of trauma. Plain x-rays revealed a lytic lesion in the upper end of the left tibia (Fig. 21). Three-phase bone imaging did not show hypervascular-

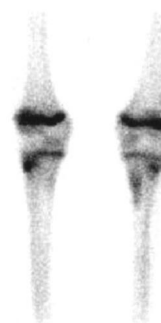


FIGURE 22. Bone scan shows mild degree of diffuse uptake in the cyst seen radiologically.



FIGURE 23. Plain radiograph shows an expansive lesion in the medial end of right clavicle.

ity corresponding to the lytic lesion seen radiologically. There was diffuse homogeneous uptake of tracer corresponding to the lytic lesion (Fig. 22). The findings favored a benign lesion such as a simple bone cyst. This was subsequently proved on histopathology.

Simple bone cyst, also called unicameral bone cyst, is of uncertain etiology and commonly presents in childhood. It frequently occurs in the proximal femur and humerus but can be seen in any other bone.²⁴ The cyst usually contains fluid unless complicated with trauma or fracture, in which case it can be contaminated with blood. When such cysts are immediately adjacent to the growth plate, they are referred to as active cysts, and when they are some distance from the growth plate, they are referred to as latent cysts.

Radiologically, a simple bone cyst appears as an area of translucency with little periosteal reaction unless complicated with fracture. Reynolds is credited with describing “the fallen fragment” sign.²⁵ Typically, the sign is identified when the

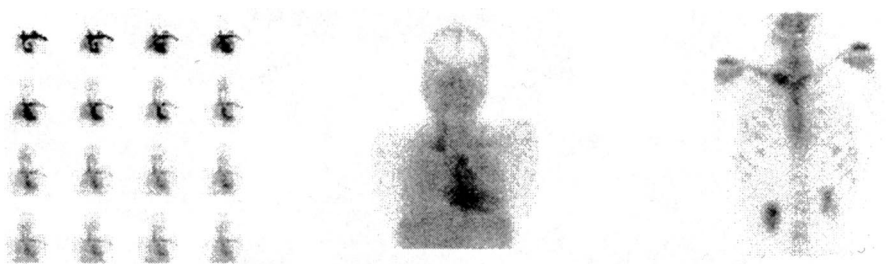


FIGURE 24. Three phase bone scan shows abnormality in all three phases suggesting neoplastic process. Histology confirmed the diagnosis of Multiple myeloma.

patient with a simple bone cyst presents with a pathologic fracture. The interior of the bone cyst could have complete or nearly complete thin bony septations within the cyst. At the time of pathologic fracture, a portion of one of these bony segments actually could break free and float to the bottom of the cyst. This is because the interior of the cyst is filled with fluid and is not a solid tumor.

Treatment options include excision of the lesion and administration of steroids. Spontaneous regression of the cysts has also been reported.²⁶

Multiple Myeloma

A 44-year-old woman developed pain in the region of the right clavicle. Plain radiographs revealed an osteolytic lesion in the medial end of the right clavicle (Fig. 23). Three phase bone scan showed a hot spot corresponding to the osteolytic lesion seen radiologically (Fig. 24). There was a history of curettage from a foot bone, which had revealed similar findings. Further work-up with serum electrophoresis revealed a myeloma band and histopathology confirmed the diagnosis of multiple myeloma.

Multiple myeloma is a plasma cell infiltrative disorder of the skeleton, which can be preceded by a solitary plasmacytoma. Initial presentation can be with local bone pain, anemia, or a pathologic fracture. Abnormal plasma cell proliferation results in raised serum proteins, predominantly globulin. In approximately half the cases, urinary Bence-Jones protein can be demonstrated.

Radiologically, the pattern of bone destruction can vary from focal areas of well-defined osteolytic lesions to diffuse osteoporosis. The bone scan can be falsely negative in multiple myeloma in view of osteolytic lesions with little bone reaction.²⁷ The sensitivity of radiography for detecting lesions in multiple myeloma has been reported to be 75% to 91%. Scintigraphy has a sensitivity of 46% to 60%.²⁸ The other conditions associated with cold areas on bone scan are metastases from renal cell carcinoma, eosinophilic granuloma, lymphoma, and postradiotherapy.^{29,30}

SUMMARY

Lesions of the skeleton of the osteolytic type are an integral part of clinical practice. Getting to the exact diagnosis of these often enigmatic lesions can be difficult and sometimes even impossible. In such a scenario, bone scintiscanning is an extremely valuable tool in the armamentarium of the nuclear medicine practice. A 3-phase bone scan done in patients with osteolytic lesions not only provides information about the vascularity of the lytic lesions, hence differentiating the benign (less likely to be vascular) from the malignant (more likely to be vascular) lesion, but is perhaps the only available test that defines bone physiology as well.

Metastatic bone diseases and metabolic bone disease can be indirectly diagnosed by a bone scan. Bone scintiscan-

ning also offers valuable leads indicating which other tests need to be performed to arrive at the ultimate diagnosis of an osteolytic lesion.

Thus, a 3-phase bone scan is crucial as well as complementary to computed tomography scanning and magnetic resonance imaging in the evaluation of osteolytic lesions.

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